

CS & IT ENGINEERING

Digital Logic



Boolean Theorems and Logical Gates

Lecture No. 4



By- Aditya sir

Recap of Previous Lecture



Topic

Problem Solving
POS & SOP

Basic Gates : AND, OR, NOT



Topics to be Covered



Topic



Practice Questions

Arithmetic Gates: + XOR, XNOR



About Aditya Jain sir



1. Appeared for GATE during BTech and secured AIR 60 in GATE in very first attempt - City topper
2. Represented college as the first Google DSC Ambassador.
3. The only student from the batch to secure an internship at Amazon. (9+ CGPA)
4. Had offer from IIT Bombay and IISc Bangalore to join the Masters program
5. Joined IIT Bombay for my 2 year Masters program, specialization in Data Science
6. Published multiple research papers in well known conferences along with the team
7. Received the prestigious excellence in Research award from IIT Bombay for my Masters thesis
8. Completed my Masters with an overall GPA of 9.36/10
9. Joined Dream11 as a Data Scientist
10. Have mentored 12,000+ students & working professions in field of Data Science and Analytics
11. Have been mentoring & teaching GATE aspirants to secure a great rank in limited time
12. Have got around 27.5K followers on LinkedIn where I share my insights and guide students and professionals.



Telegram



Telegram Link for Aditya Jain sir: https://t.me/AdityaSir_PW

Basic gates

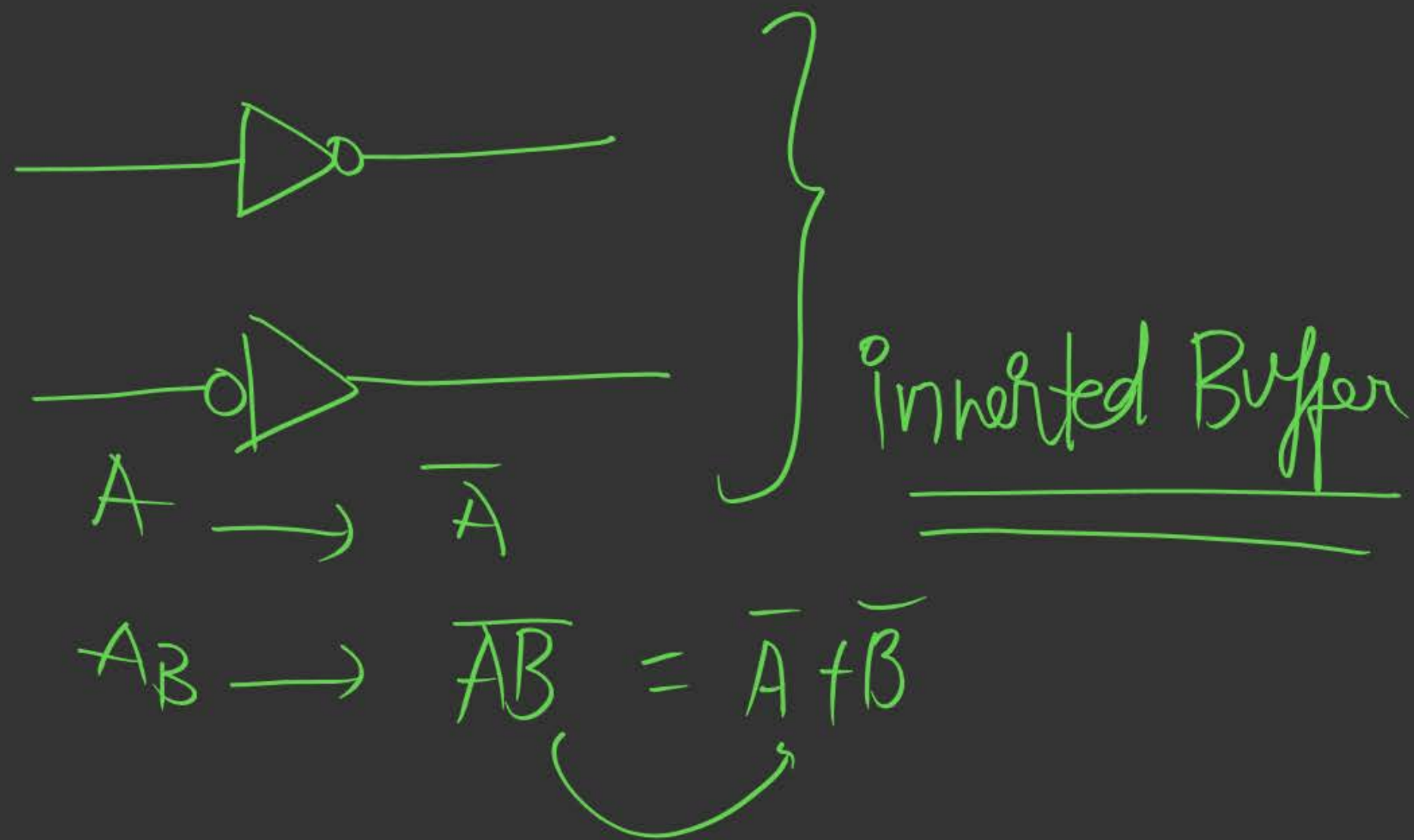
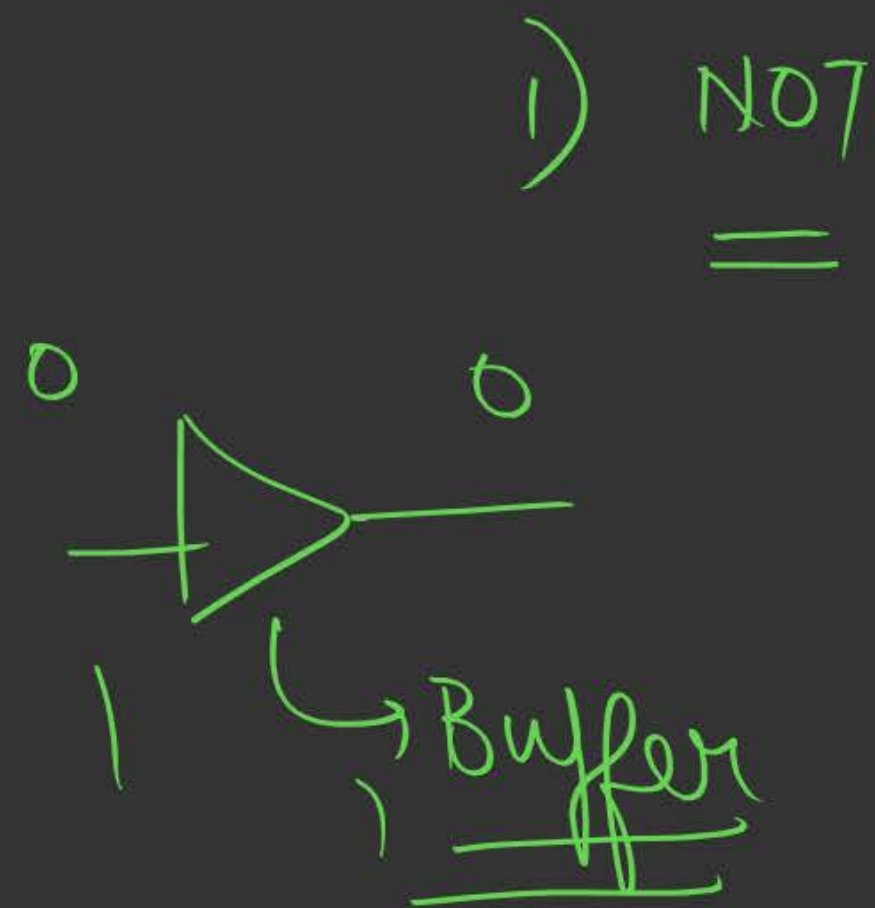
- 1) AND ✓
- 2) OR ✓
- 3) NOT ✓

Arithmetic gates

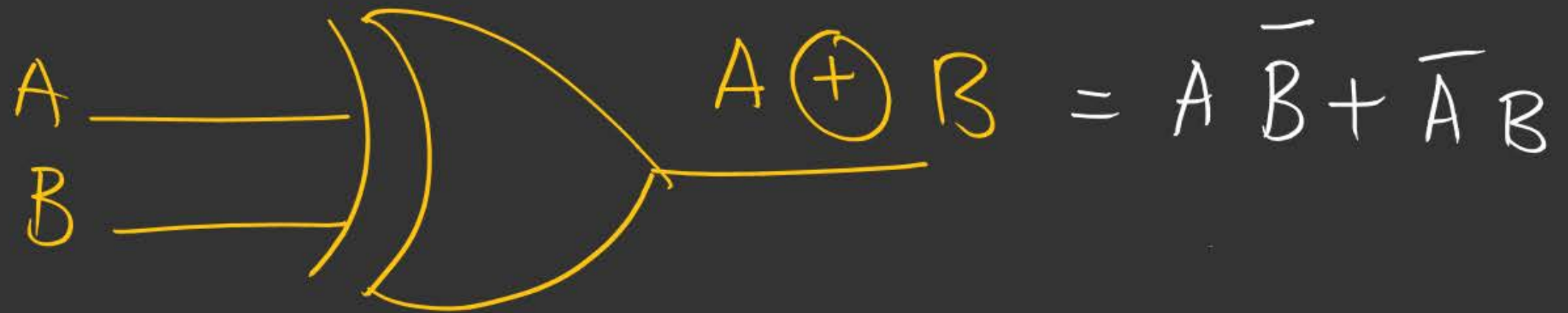
- 1) XOR
- 2) XNOR

Universal gates

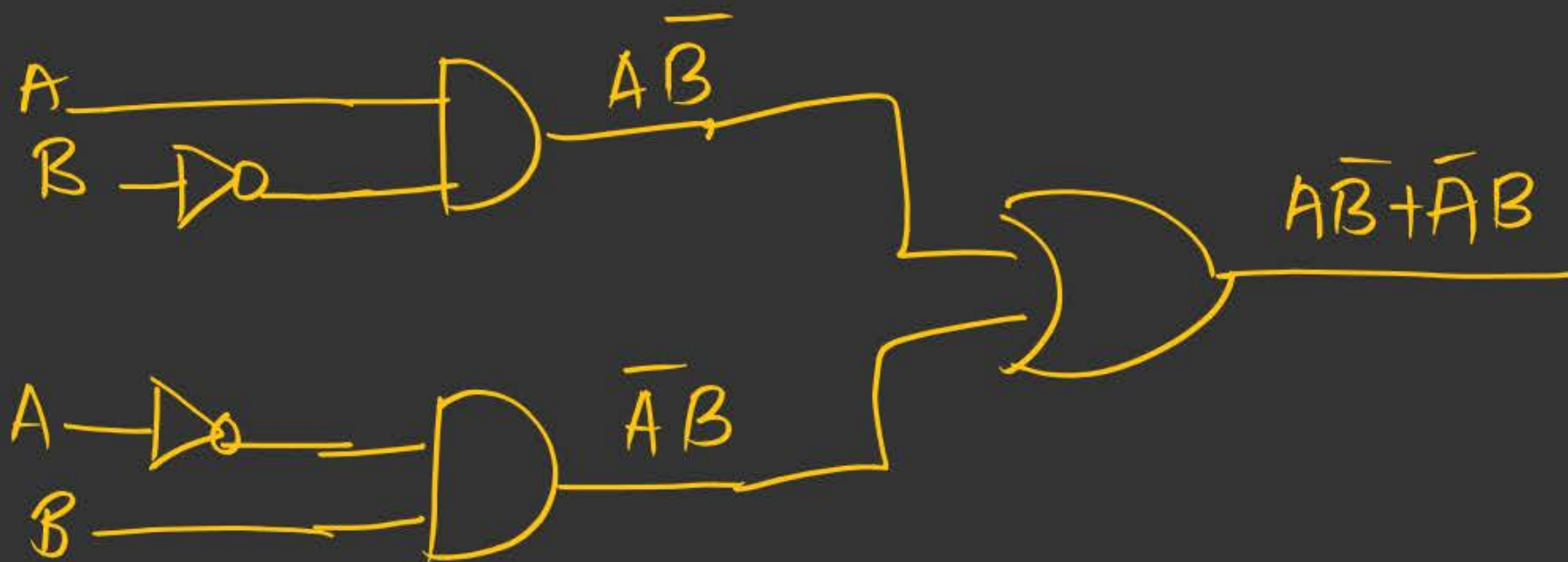
- 1) NAND
- 2) NOR



➤ XOR → Exclusive 'OR'



Representation:



$$A \oplus B = A \bar{B} + \bar{A} B = \bar{A} B + A \bar{B}$$

XOR

Truth table

	A	B	A ⊕ B
0	0	0	0
1 →	0	1	1
2 →	1	0	1
3	1	1	0

$$A \oplus B = \bar{A}B + A\bar{B}$$

☆

A	B
0	0
0	1
1	0
1	1

0 → $\bar{0} \cdot 0 + 0 \cdot \bar{0} = 1 \cdot 0 + 0 \cdot 1 = 0 + 0 = 0$

0 → $\bar{0} \cdot 1 + 0 \cdot \bar{1} = 1 \cdot 1 + 0 \cdot 0 = 1 + 0 = 1$

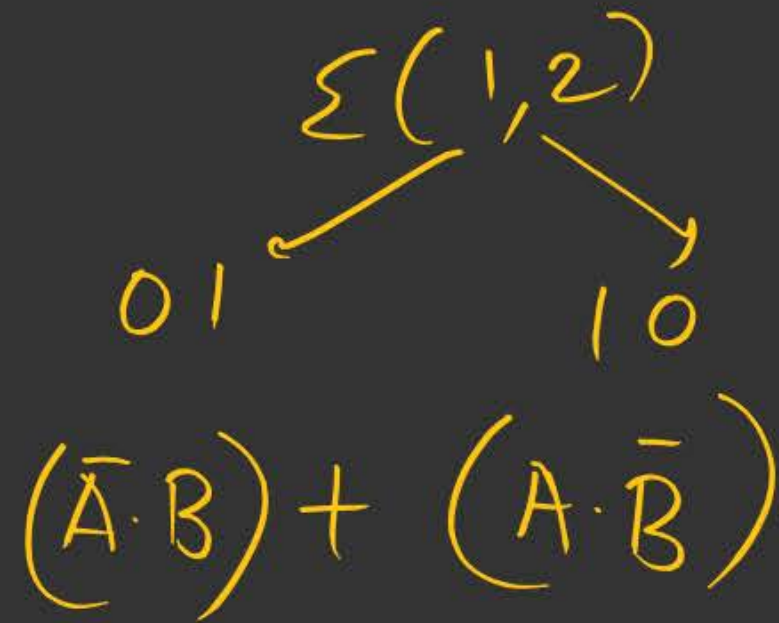
1 → $\bar{1} \cdot 0 + 1 \cdot \bar{0} = 0 \cdot 0 + 1 \cdot 1 = 0 + 1 = 1$

1 → $\bar{1} \cdot 1 + 1 \cdot \bar{1} = 0 \cdot 1 + 1 \cdot 0 = 0 + 0 = 0$

$A=1$ SOP (1s)

$\overline{A}=0$

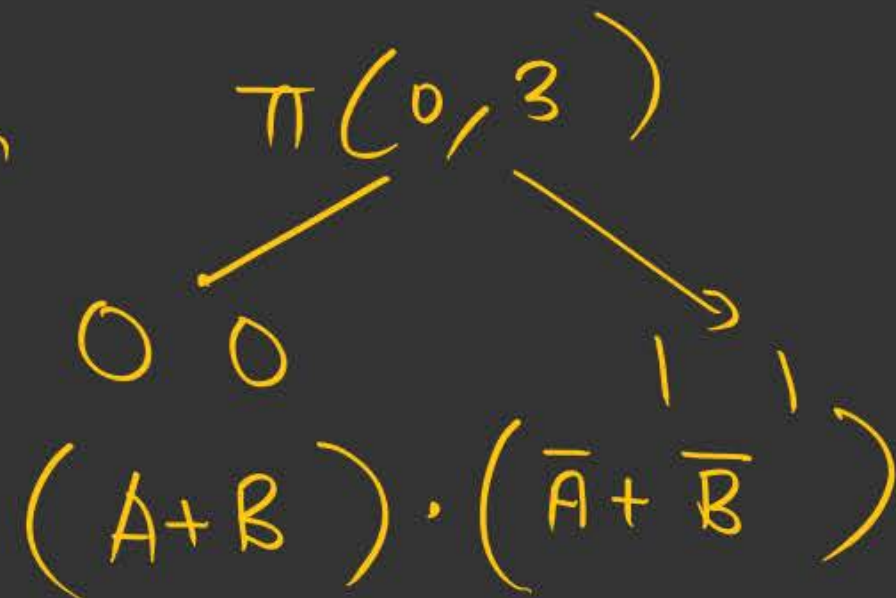
$\Sigma(1,2)$



$A=0$ POS $(0,3)$

$\bar{A}=1$

$\Pi(0,3)$



$$(A+B) \cdot (\bar{A}+\bar{B}) = \cancel{A \cdot \bar{A}}^0 + A\bar{B} + B\bar{A} + \cancel{B \cdot \bar{B}}^0$$

$$= A\bar{B} + B\bar{A} \rightarrow \underline{\underline{\text{SOP}}}$$

Trick Remember:

SOP:

$$A \oplus B = \bar{A} \cdot B + A \cdot \bar{B}$$

POS

$$A \oplus B = (A + B) \cdot (\bar{A} + \bar{B})$$

Practice

$$1) \quad \overline{A} \oplus B = \overline{A} \cdot B + \overline{\overline{A}} \cdot \overline{B} = A \cdot B + \overline{A} \cdot \overline{B}$$

$$2) \quad A \oplus \overline{B} = \overline{A} \cdot \overline{B} + A \cdot \overline{\overline{B}} = \overline{A} \cdot \overline{B} + A \cdot B$$

$$3) \quad \underline{\underline{\overline{A} \oplus \overline{B}}} = \overline{\overline{A}} \cdot \overline{\overline{B}} + \overline{\overline{A}} \cdot \overline{\overline{B}} = A \cdot \overline{B} + \overline{A} \cdot B$$

$= A \oplus B$

✓ 1) $AB \oplus BC$

✓ 2) $P \oplus QP =$



$$AB \oplus BC = \overline{AB} \cdot BC + AB \cdot \overline{BC}$$

$$= (\overline{A} + \overline{B}) \cdot BC + AB \cdot (\overline{B} + \overline{C})$$

$$= \overline{A} \cdot BC + \overline{B} \cdot BC + AB \cdot \overline{B} + AB \cdot \overline{C}$$

$$= \overline{A} \cdot BC + AB \cdot \overline{C}$$

$$= B \cdot (\overline{A}C + A\overline{C}) = B \cdot ((A+C)(\overline{A}+\overline{C}))$$

$$= B \cdot (A \oplus C)$$

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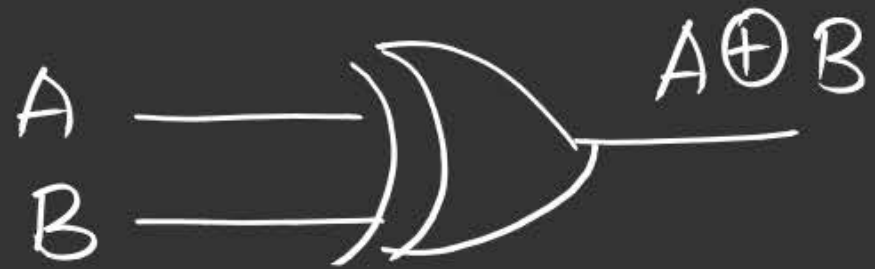
$$2) P \oplus QP = \bar{P} \cdot QP + P \cdot \overline{QP}$$

$$= P \cdot \overline{QP}$$

$$= P(\bar{Q} + \bar{P})$$

$$= P\bar{Q} + \cancel{P\bar{P}} = P\bar{Q}$$

1) Commutative Law.



$\equiv$



$$A \oplus B = \bar{A} \cdot B + A \cdot \bar{B}$$

$$B \oplus A = \bar{B} \cdot A + B \cdot \bar{A}$$

$$= \bar{A} B + A \bar{B}$$

2) Associative Law:

→ Multiple i/p XOR can be designed using
2 i/p XOR.

$$A \oplus B \oplus C \Rightarrow (A \oplus B) = P \Rightarrow P \oplus C$$

$$\Rightarrow (B \oplus C) = Q \Rightarrow A \oplus Q$$

$$A \oplus B \oplus C = B \oplus C \oplus A$$

→ associative law holds

★ Properties of XOR :

$$1) A \oplus A = 0$$

$$\hookrightarrow A \oplus A = \overline{A} \cdot A + A \cdot \overline{A} = 0 + 0 = 0$$

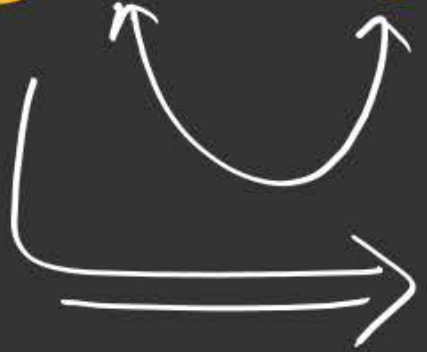
$$2) \overline{A} \oplus \overline{A} = 0$$

$$3) A \oplus \bar{A} = 1$$

$$\hookrightarrow \bar{A} \cdot \bar{A} + A \cdot \bar{\bar{A}} = \bar{A} + A = 1$$

$$4) \bar{A} \oplus A = 1$$

$$5) A \oplus A = 0$$



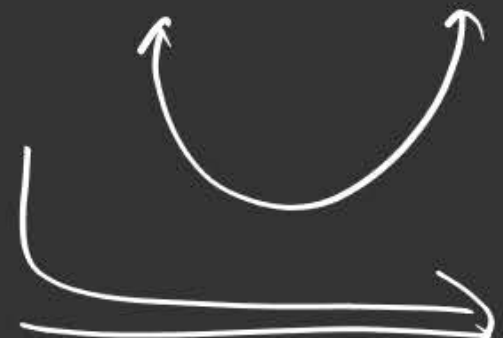
$$A \oplus 0 = A$$

Proof

$$\begin{aligned} A \oplus 0 &= \bar{A} \cdot 0 + A \cdot \bar{0} \\ &= 0 + A \cdot 1 \\ &= A \end{aligned}$$

Anything $\oplus 0 =$ itself

$$6) A \oplus \bar{A} = 1$$



$$A \oplus 1 = \bar{A}$$

Proof -

$$A \oplus 1 = \bar{A} \cdot 1 + A \cdot \bar{1}$$

$$= \bar{A} + A \cdot 0$$

$$= \bar{A}$$

anything $\oplus 1 = \text{complement}$

$$\overset{x}{\sim} PQ \oplus 0 = PQ$$

$$PQ \oplus 1 = \overline{PQ} = \bar{P} + \bar{Q}$$

★ Exchange Property of XOR:

Let $A \oplus B = C$ (given)



$A \oplus C = B$

Proof:

given $(A \oplus B) = C$ — (1)

$A \oplus C$ $= A \oplus (A \oplus B) = (A \oplus A) \oplus B$

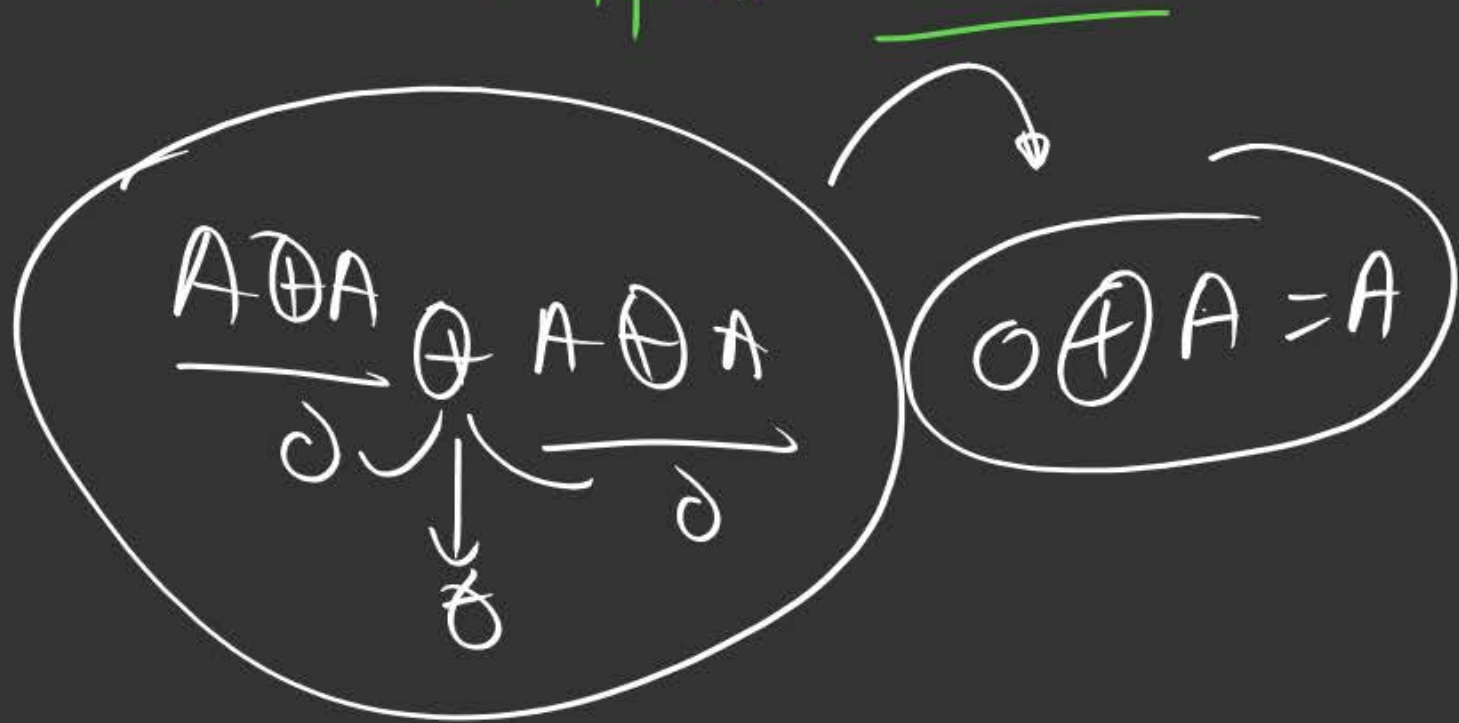
$= 0 \oplus B = B$

$$A \oplus B = C \Rightarrow A \oplus C = B$$

$$A \oplus B = C \Rightarrow B \oplus A = C \Rightarrow B \oplus C = A$$

$$A \oplus A \oplus A \oplus A \dots n \text{ times}$$

then O/p is



$$\text{odd} = \text{even} + 1$$

$$\text{if } n = \text{even}, \Rightarrow 0$$

$$\text{if } n = \text{odd} \Rightarrow A$$

eg - $A \oplus A = 0$

$$\underbrace{A \oplus A}_{0} \oplus \underbrace{A \oplus A}_{0}$$

$$0 \oplus 0 = 0$$

$$\underbrace{A \oplus A}_{0} \oplus A = A \quad \checkmark$$

Buffer & Inverting Buffer:



control = 0
= 1 } enable

$$\left. \begin{array}{l} \text{Control} = 0, A \oplus 0 = A \\ \quad \quad \quad = 1, A \oplus 1 = \bar{A} \end{array} \right\} \underline{\text{enable}}$$

$$\text{Control} = 0, A \oplus 0 = A \longrightarrow \text{Buffer}$$

$$\text{Control} = 1, A \oplus 1 = \bar{A} \longrightarrow \text{Inverting Buffer}$$

Practice:-

$$PQ + P\bar{Q}R$$

simplifies to

A) $PQ + QR$

☒ B) $PQ + PR$

C) $PQ + \bar{Q}R$

D) PQR

$$PQ + P\bar{Q}R$$

$$\star = P[Q + \bar{Q}R]$$

$$\star = P[\underbrace{(Q + \bar{Q})}_{1} \cdot (Q + R)] = P[(Q + R)] = P \cdot Q + P \cdot R$$

(Q) $PQ + Q\bar{R} + \bar{P}R$ simplifies to?

- A) $(\bar{P} + Q)(Q + R)$
- B) $PQ + Q\bar{R}$
- C) $PQ + \bar{P}R$
- D) $\bar{P}R + Q\bar{R}$

H.W

(Q3) $f(P, Q, R) = \overline{\Sigma(1, 2, 4, 6, 7)}$

MSQ

then $f(P, Q, R)$ is?

A) $\Pi(0, 3, 5)$

C) $\Sigma(1, 2, 4, 6, 7)$

B) $\Sigma(0, 3, 5)$

D) $\Pi(1, 2, 4, 6, 7)$

$$(Q4) \quad (P+Q+R \cdot S) (\bar{P}+Q+\bar{R} \cdot S) (\bar{P}+Q+R)$$

simplify



2 mins Summary



Topic

Topic

Revision

Topic

XOR

Topic

Practice Quesns

Topic



THANK - YOU